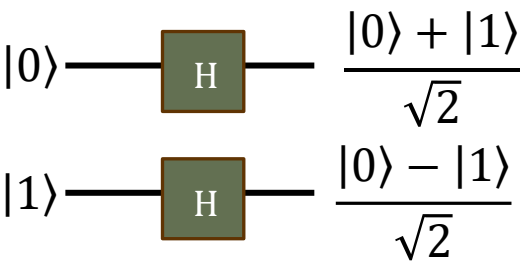


Introduction to Quantum Computing (L-6)

By: Dr. Samik Mukherjee (Ph.D.)
Assistant Professor, Jio Institute

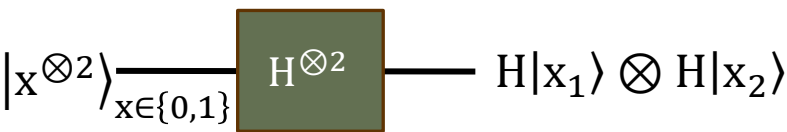
Hadamard Gate – General Expression



$$H|x\rangle_{x \in \{0,1\}} = \frac{|0\rangle + (-1)^x |1\rangle}{\sqrt{2}}$$

$(-1)^0 = 1$ and $(-1)^1 = -1$

Hadamard gate on 2-qubit register



$$\frac{1}{\sqrt{2}} \sum_{z_1 \in \{0,1\}} (-1)^{x_1 \cdot z_1} |z_1\rangle \otimes \frac{1}{\sqrt{2}} \sum_{z_2 \in \{0,1\}} (-1)^{x_2 \cdot z_2} |z_2\rangle = \frac{1}{\sqrt{2^2}} \sum_{z \in \{00,01,10,11\}} (-1)^{x \cdot z} |z\rangle$$

General form for single qubit

$$H|x\rangle = \frac{1}{\sqrt{2}} \sum_{z \in \{0,1\}} (-1)^{x \cdot z} |z\rangle$$

Hadamard Gate – General Expression

Hadamard gate on 2-qubit register

$|x^{\otimes 2}\rangle_{x \in \{0,1\}}$  $H|x_1\rangle \otimes H|x_2\rangle$

$$\frac{1}{\sqrt{2}} \sum_{z_1 \in \{0,1\}} (-1)^{x_1 \cdot z_1} |z_1\rangle \otimes \frac{1}{\sqrt{2}} \sum_{z_2 \in \{0,1\}} (-1)^{x_2 \cdot z_2} |z_2\rangle = \frac{1}{\sqrt{2^2}} \sum_{z \in \{00,01,10,11\}} (-1)^{x \cdot z} |z\rangle$$

$|x^{\otimes 2}\rangle = |00\rangle$

z	x · z = x₁z₁ + x₂z₂	(−1)^{x·z}
00	0	1
01	0	1
10	0	1
11	0	1

$$H^{\otimes 2}|00\rangle = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{\sqrt{2^2}}$$

$|x^{\otimes 2}\rangle = |01\rangle$

z	x · z = x₁z₁ + x₂z₂	(−1)^{x·z}
00	0	1
01	1	−1
10	0	1
11	1	−1

$$H^{\otimes 2}|00\rangle = \frac{|00\rangle - |01\rangle + |10\rangle - |11\rangle}{\sqrt{2^2}}$$

$|x^{\otimes 2}\rangle = |11\rangle$

z	x · z = x₁z₁ + x₂z₂	(−1)^{x·z}
00	0	1
01	1	−1
10	0	−1
11	1	1

$$H^{\otimes 2}|00\rangle = \frac{|00\rangle - |01\rangle - |10\rangle + |11\rangle}{\sqrt{2^2}}$$

Hadamard Gate – General Expression

Hadamard gate on n-qubit register

$$|x^{\otimes n}\rangle_{x \in \{0,1\}} \xrightarrow{H^{\otimes n}} H|x_1\rangle \otimes H|x_2\rangle \dots \otimes H|x_n\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} (-1)^{x \cdot z} |z\rangle$$

$|x^{\otimes 3}\rangle = |101\rangle$

z	x · z = x₁z₁ + x₂z₂ + x₃z₃	(−1)^{x·z}
000	0	1
001	1	−1
010	0	1
011	1	−1
100	1	−1
101	2	+1
110	1	−1
111	2	+1

$$H^{\otimes 3}|101\rangle = \frac{|000\rangle - |001\rangle + |010\rangle - |011\rangle + |100\rangle - |101\rangle + |110\rangle - |111\rangle}{\sqrt{8}}$$

$$H^{\otimes 3}|111\rangle = \frac{|000\rangle - |001\rangle - |010\rangle + |011\rangle - |100\rangle + |101\rangle + |110\rangle - |111\rangle}{\sqrt{8}}$$

Hadamard Gate – General Expression

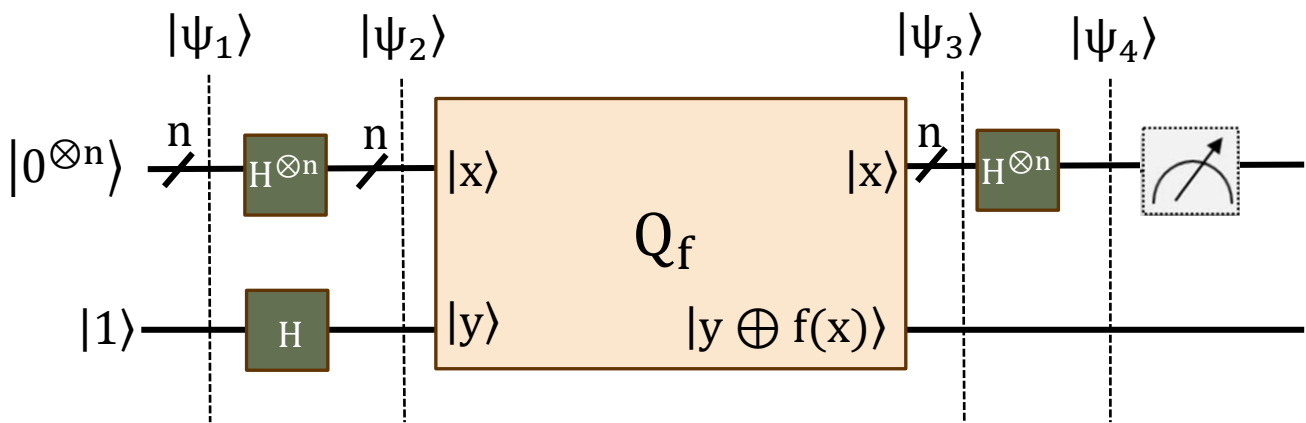
Hadamard gate on a superposition of states

$$\begin{aligned} & H^{\otimes 3} \frac{|000\rangle + |110\rangle - |100\rangle - |111\rangle}{\sqrt{8}} \\ &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \frac{|0\rangle + |1\rangle}{\sqrt{2}} \frac{|0\rangle + |1\rangle}{\sqrt{2}} + \frac{|0\rangle - |1\rangle}{\sqrt{2}} \frac{|0\rangle - |1\rangle}{\sqrt{2}} \frac{|0\rangle + |1\rangle}{\sqrt{2}} - \frac{|0\rangle - |1\rangle}{\sqrt{2}} \frac{|0\rangle + |1\rangle}{\sqrt{2}} \frac{|0\rangle + |1\rangle}{\sqrt{2}} - \frac{|0\rangle - |1\rangle}{\sqrt{2}} \frac{|0\rangle - |1\rangle}{\sqrt{2}} \frac{|0\rangle - |1\rangle}{\sqrt{2}} \end{aligned}$$

	000	001	010	011	100	101	110	111
000	+	+	+	+	+	+	+	+
110	+	+	−	−	−	−	+	+
−100	−	−	−	−	+	+	+	+
−111	−	+	+	−	+	−	−	+
	0 000⟩	2 001⟩	0 010⟩	−2 011⟩	2 100⟩	0 010⟩	2 110⟩	4 111⟩

$$\frac{1}{\sqrt{2^3}} \sum_{z \in \{0,1\}^3} (-1)^{x \cdot z} |z\rangle = \frac{2|001\rangle - 2|011\rangle + 2|100\rangle + 2|110\rangle + 4|111\rangle}{2\sqrt{2^3}}$$

Recap - Deutsch Jozsa Algorithm



Constant	
Input	f(x)
000	1
001	1
010	1
011	1
100	1
101	1
110	1
111	1

Balanced	
Input	f(x)
000	0
001	0
010	1
011	1
100	1
101	1
110	0
111	0

Determine whether a Boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$ is **constant or balanced**

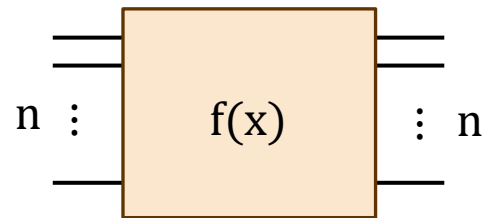
Classical Algorithms: In worst case, need $2^{n-1} + 1$ evaluations
Complexity: $\mathcal{O}(2^n)$

Quantum algorithm uses superposition + interference. Only one oracle query is needed.
Complexity reduces from exponential $\rightarrow \mathcal{O}(1)$

$$H^{\otimes n} \rightarrow Q_f \rightarrow H^{\otimes n} \rightarrow \text{Measurement}$$

If we measure all 0s in the first register $\rightarrow$ Function is constant
If we measure anything else $\rightarrow$ Function is balanced

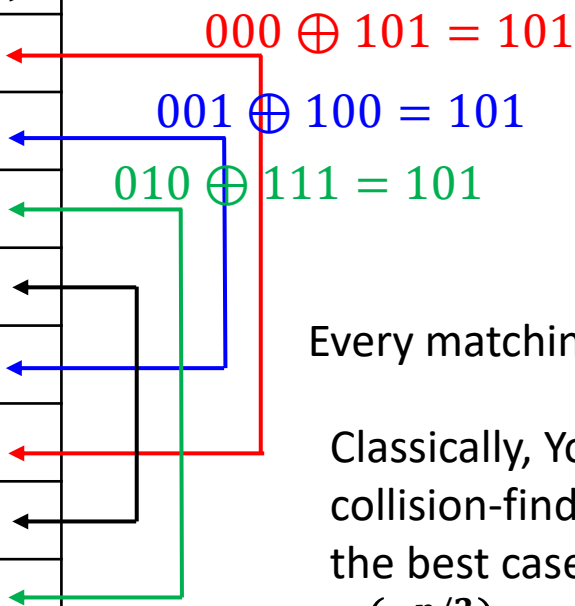
Simon Algorithm



Given a 2-1 function $f(x): \{0,1\}^n \rightarrow \{0,1\}^n$, such that $f(x) = f(x \oplus s)$ for $s \in \{0,1\}^n$ and $s \neq 0$, our goal is to find the secret bit string 's'

Let $n = 3$ and let our secret bit string be $s = 101$

Domain (x)	Range f(x)
000	111
001	000
010	110
011	010
100	000
101	111
110	010
111	110



Look at the table carefully. You will notice:

- Every output appears **exactly twice**
- Each repeated output corresponds to **two different inputs**

For example:

$$\begin{aligned} f(000) &= f(101) = 111 \\ f(001) &= f(100) = 000 \\ f(010) &= f(111) = 110 \\ f(011) &= f(110) = 010 \end{aligned}$$

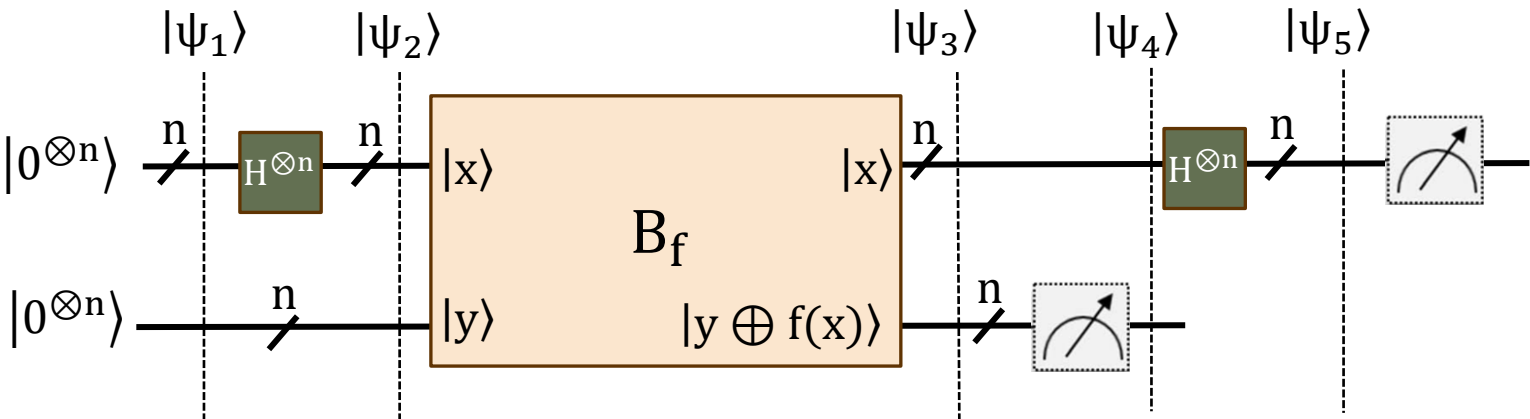
This is what **2-to-1** means.

Every matching pair gives the same XOR result. That XOR value is the secret string **101**

Classically, You must keep querying the oracle until you see a collision. This is a collision-finding problem. In the worst case we need to make $O(2^n)$ oracle calls. In the best case, with the best randomized classical strategy we still need to make $O(2^{n/2})$ oracle calls

Simon's algorithm: Uses superposition and needs only $O(n)$ oracle queries. This leads to an exponential speedup

Simon Algorithm



Lets take n = 2

x	f(x)
00	01
01	11
10	01
11	11

$$s = 00 \oplus 10 = 10$$

$$|\psi_1\rangle = |00\rangle|00\rangle$$

$$|\psi_2\rangle = H^{\otimes 2}|00\rangle|00\rangle$$

$$|\psi_2\rangle = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2} |00\rangle$$

- First register: uniform superposition
- Second register: still zero

After the Oracle B_f

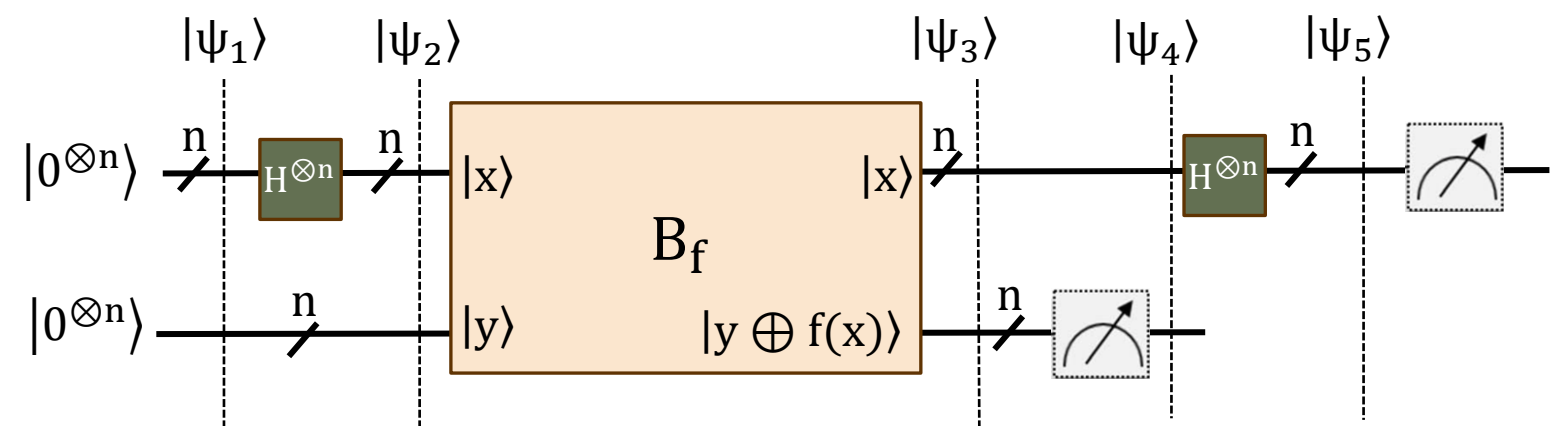
- Recall: $B_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$
- In this case, $|y\rangle = |00\rangle$ which means $B_f|x\rangle|y\rangle = |x\rangle|f(x)\rangle$

$$|\psi_3\rangle = \frac{|00\rangle|01\rangle + |10\rangle|01\rangle + |01\rangle|11\rangle + |11\rangle|11\rangle}{2}$$

To get to $|\psi_4\rangle$ lets measure the 2nd register. What measuring the 2nd register does is it

- Picks one value
- Deletes all incompatible terms
- Renormalizes what remains

Simon Algorithm



x	f(x)
00	01
01	11
10	01
11	11

$$|\psi_3\rangle = \frac{|00\rangle|01\rangle + |10\rangle|01\rangle + |01\rangle|11\rangle + |11\rangle|11\rangle}{2}$$
$$= \frac{1}{2}\{(|00\rangle + |10\rangle)|01\rangle + (|01\rangle + |11\rangle)|11\rangle\}$$

To get to $|\psi_3\rangle$ let's measure the 2nd register.
What measuring the 2nd register does is it

- Picks one value
- Deletes all incompatible terms
- Renormalizes what remains

The 2nd register can either measure 01 or it can measure 11

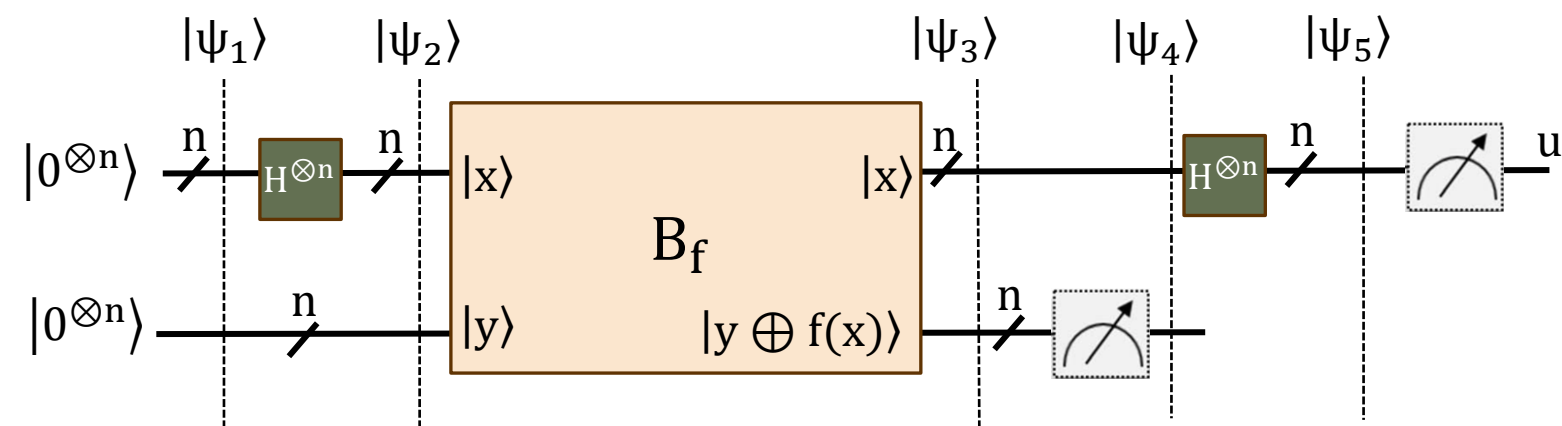
Case A: Say the 2nd register measures 01
When the 2nd register measures $f(z) = 01$,
the first register must collapse to $\frac{|00\rangle + |10\rangle}{\sqrt{2}}$

Case B: Say the 2nd register measures 11
When the 2nd register measures $f(z) = 11$,
the first register must collapse to $\frac{|01\rangle + |11\rangle}{\sqrt{2}}$

If the secret message is 's', then measuring $f(z)$ in the 2nd register forces the state of the 1st register to become $\frac{|z\rangle + |z \oplus s\rangle}{\sqrt{2}}$

Measuring the second register “reveals” the hidden pairing without telling us ‘s’.

Simon Algorithm



x	f(x)
00	01
01	11
10	01
11	11

Case A: Say the 2nd register measures 01
1st register collapsed to $\frac{|00\rangle + |10\rangle}{\sqrt{2}}$

$$|\psi_5\rangle = H^{\otimes 2} \frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

$$H^{\otimes 2}|00\rangle = \frac{1}{2} \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{\sqrt{2}}$$

$$H^{\otimes 2}|10\rangle = \frac{1}{2} \frac{|00\rangle + |01\rangle - |10\rangle - |11\rangle}{\sqrt{2}}$$

$$|\psi_5\rangle = \frac{2|00\rangle + 2|01\rangle}{2\sqrt{2}} = \frac{|00\rangle + |01\rangle}{\sqrt{2}}$$

Next, we measure the 1st register. The two possible outcomes of measurement are $u = 00$ or $u = 01$.

Simons algorithm guarantees that $u \cdot s = 0 \pmod{2}$

If $u = (u_1, u_2)$ and $s = (s_1, s_2)$ then $u_1 s_1 \oplus u_2 s_2 = 0$

Lets say during the 1st run you get $u = (01)$. $u_1 = 0, u_2 = 1$ } $s = (s_1, s_2) = (s_1, 0)$
 $0 \cdot s_1 \oplus 1 \cdot s_2 = 0 \Rightarrow 0 \oplus s_2 = 0 \Rightarrow s_2 = 0$
Lets say during the 2nd or the 3rd or the 4th run you get $u = (00)$ } $s \neq (00)$
 $0 \cdot s_1 \oplus 0 \cdot s_2 = 0 \Rightarrow 0 \oplus 0 = 0 \Rightarrow$ No information } s must be (10)

Mod (n) - Arithmetic

Mod 2 means: take the remainder after division by 2.

So, the only possible answers are: 0 (even number) or 1 (odd number)

Examples:

- $5 \bmod 2 = 1$
- $8 \bmod 2 = 0$
- $11 \bmod 2 = 1$

Mod 3 means: remainder after division by 3.

Possible answers: 0, 1, 2

Examples:

- $7 \bmod 3 = 1$
- $10 \bmod 3 = 1$
- $14 \bmod 3 = 2$

a	b	a + b	(a + b) mod 2	a $\oplus$ b
0	0	0	0	0
0	1	1	1	1
1	0	1	1	1
1	1	2	0	0

Addition in Mod2 is equivalent to XOR

$$y = y_1y_2y_3 = 101 \ \& \ s = s_1s_2s_3 = 110$$

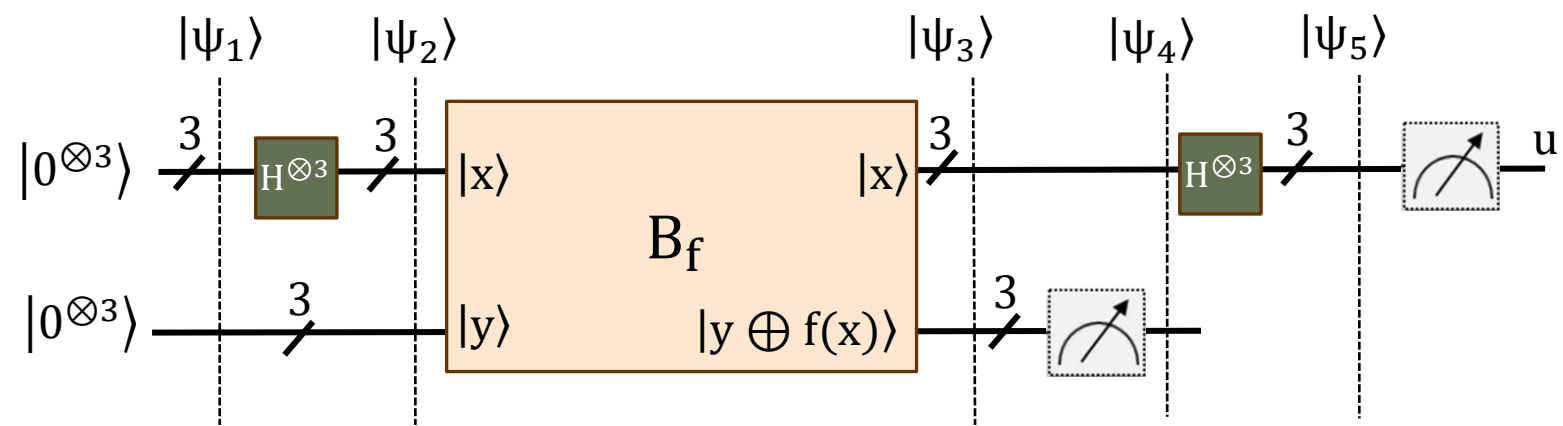
$$y.s = (1.1) + (0.1) + (1.0) = 1 \pmod{2} = 1$$

$$y = y_1y_2y_3 = 111 \ \& \ s = s_1s_2s_3 = 101$$

$$y.s = (1.1) + (0.1) + (1.1) = 2 \pmod{2} = 0$$

Simon Algorithm

x	f(x)
000	010
001	111
010	001
011	100
100	111
101	010
110	100
111	001



$|\psi_1\rangle = |000\rangle|000\rangle$

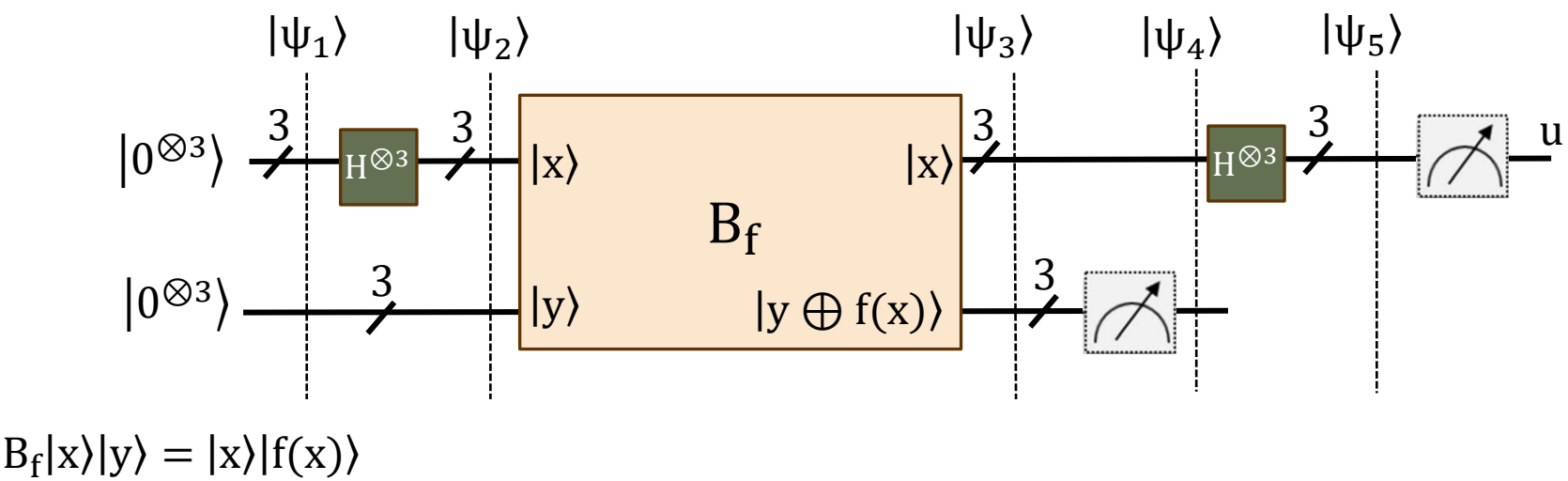
$|\psi_2\rangle = H^{\otimes 3}|000\rangle|000\rangle = \frac{1}{\sqrt{2^3}} \sum_{z \in \{0,1\}^3} (-1)^{x \cdot z} |z\rangle|000\rangle = \frac{1}{\sqrt{2^3}} \sum_{z \in \{0,1\}^3} |z\rangle|000\rangle$

$|\psi_2\rangle = \frac{|000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle + |101\rangle + |110\rangle + |111\rangle}{\sqrt{2^3}} |000\rangle$

- Recall: $B_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$
- Since, $|y\rangle = |000\rangle$ means $B_f|x\rangle|y\rangle = |x\rangle|f(x)\rangle$

Simon Algorithm

x	f(x)
000	010
001	111
010	001
011	100
100	111
101	010
110	100
111	001

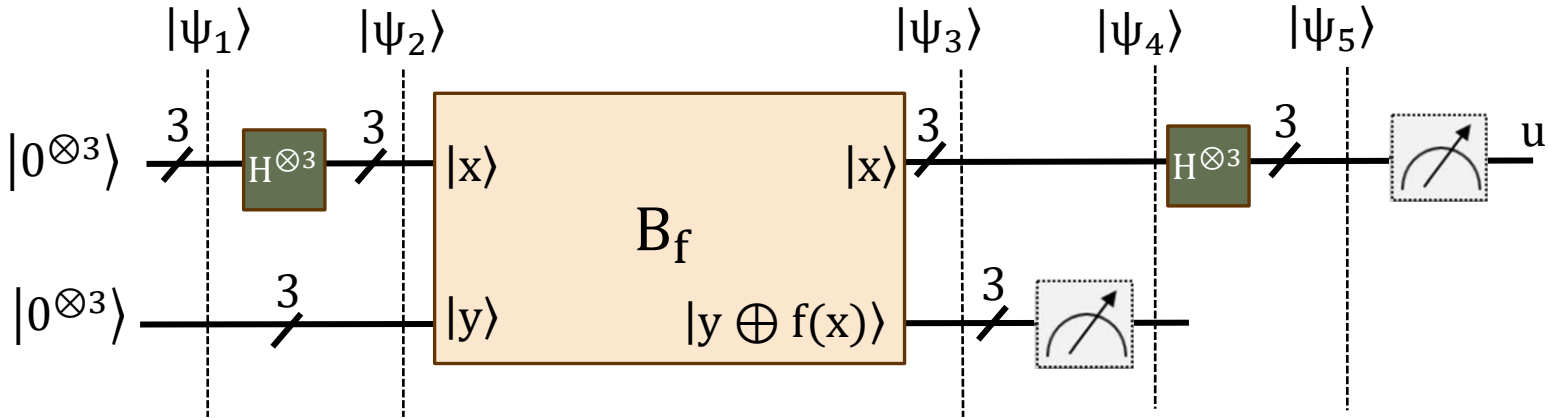


$$|\psi_3\rangle = \frac{|000\rangle|010\rangle + |001\rangle|111\rangle + |010\rangle|001\rangle + |011\rangle|100\rangle + |100\rangle|111\rangle + |101\rangle|010\rangle + |110\rangle|100\rangle + |111\rangle|001\rangle}{\sqrt{2^3}}$$

Let's assume that during the 1st run of the circuit, the second register measures $|010\rangle$. We call this $f(z)$. The moment this happens, the first register state collapses to $|\psi_4\rangle = \frac{|000\rangle + |101\rangle}{\sqrt{2}}$

Simon Algorithm

x	f(x)
000	010
001	111
010	001
011	100
100	111
101	010
110	100
111	001



$$|\psi_4\rangle = \frac{|000\rangle + |101\rangle}{\sqrt{2}}$$

$$|\psi_5\rangle = H^{\otimes 3} \frac{|z\rangle + |z \oplus s\rangle}{\sqrt{2}}$$

We do not know $|\psi_4\rangle$ since we have not measured it yet. All we know that if the second register have measured $f(z)$, then $|\psi_4\rangle$ must be in the state $\frac{|z\rangle + |z \oplus s\rangle}{\sqrt{2}}$

$$H^{\otimes 3}|z\rangle = \frac{1}{\sqrt{2^3}} \sum_{u \in \{0,1\}^3} (-1)^{z \cdot u} |u\rangle$$

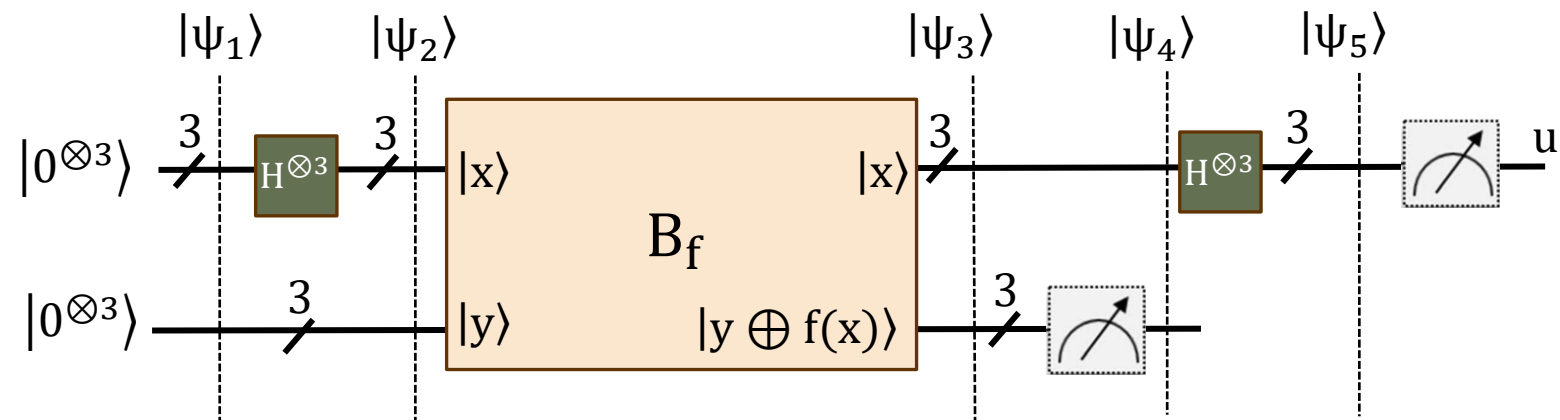
$$H^{\otimes 3}|z \oplus s\rangle = \frac{1}{\sqrt{2^3}} \sum_{u \in \{0,1\}^3} (-1)^{(z \oplus s) \cdot u} |u\rangle$$

$$\{(-1)^{z \cdot u} + (-1)^{(z \oplus s) \cdot u}\} = (-1)^{z \cdot u} \{1 + (-1)^{u \cdot s}\}$$

$$|\psi_5\rangle = \frac{1}{\sqrt{2^4}} \sum_{u \in \{0,1\}^3} \{(-1)^{z \cdot u} + (-1)^{(z \oplus s) \cdot u}\} |u\rangle$$

$|\psi_5\rangle$ will collapse to $|000\rangle$ with 100% probability if $1 + (-1)^{u \cdot s} = 0$. $|\psi_5\rangle$ will survive iff $1 + (-1)^{u \cdot s} \neq 0$. This will only happen if $u \cdot s = 0 \pmod{2}$

Simon Algorithm



$$|\psi_5\rangle = H^{\otimes 3} \frac{|000\rangle + |010\rangle}{\sqrt{2}} = \frac{|000\rangle + |010\rangle + |101\rangle + |111\rangle}{2}$$

Now, when we measure, we are equally likely to get any one of these four possible outcomes. If we get 000 we are not in luck. This will lead to $0s_1 \oplus 0 \cdot s_2 \oplus 0 \cdot s_3 = 0 \Rightarrow 0 = 0$. Lets say we measure $u = 010$

$$0s_1 \oplus 1 \cdot s_2 \oplus 0 \cdot s_3 = 0 \Rightarrow s_2 = 0$$

In the 1st run of the circuit, the second register measures $|010\rangle$. Let's assume that during the 2nd run of the circuit, the second register measures $|111\rangle$. We call this $f(z)$. The moment this happens, the first register state collapses to

$$|\psi_4\rangle = \frac{|001\rangle + |100\rangle}{\sqrt{2}}$$

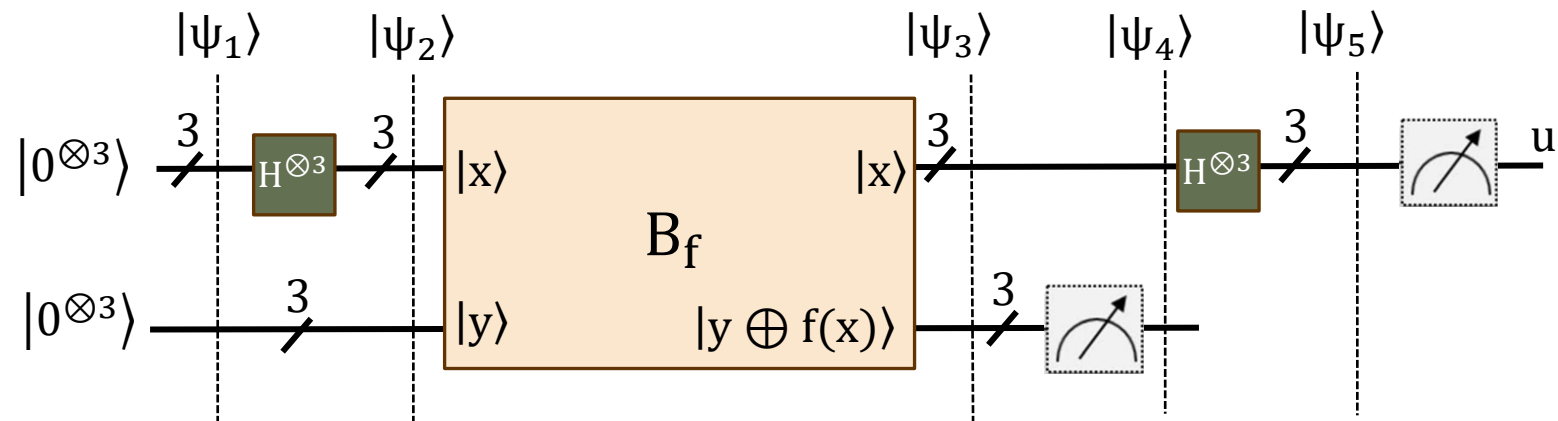
We measure the circuit again.

$$|\psi_3\rangle = \frac{|000\rangle|010\rangle + |001\rangle|111\rangle + |010\rangle|001\rangle + |011\rangle|100\rangle + |100\rangle|111\rangle + |101\rangle|010\rangle + |110\rangle|100\rangle + |111\rangle|001\rangle}{\sqrt{2^3}}$$

Simon Algorithm

$$|\psi_4\rangle = \frac{|001\rangle + |101\rangle}{\sqrt{2}}$$

$$|\psi_5\rangle = H^{\otimes 3} \frac{|001\rangle + |110\rangle}{\sqrt{2}} = \frac{|000\rangle - |011\rangle - |101\rangle + |110\rangle}{2}$$



Now, when we measure, we are equally likely to get any one of these four possible outcomes. If we get 000 we are not in luck. Lets say we measure $u = 101$

$$1s_1 \oplus 0 \cdot s_2 \oplus 1 \cdot s_3 = 0 \quad \Rightarrow s_1 = s_3$$

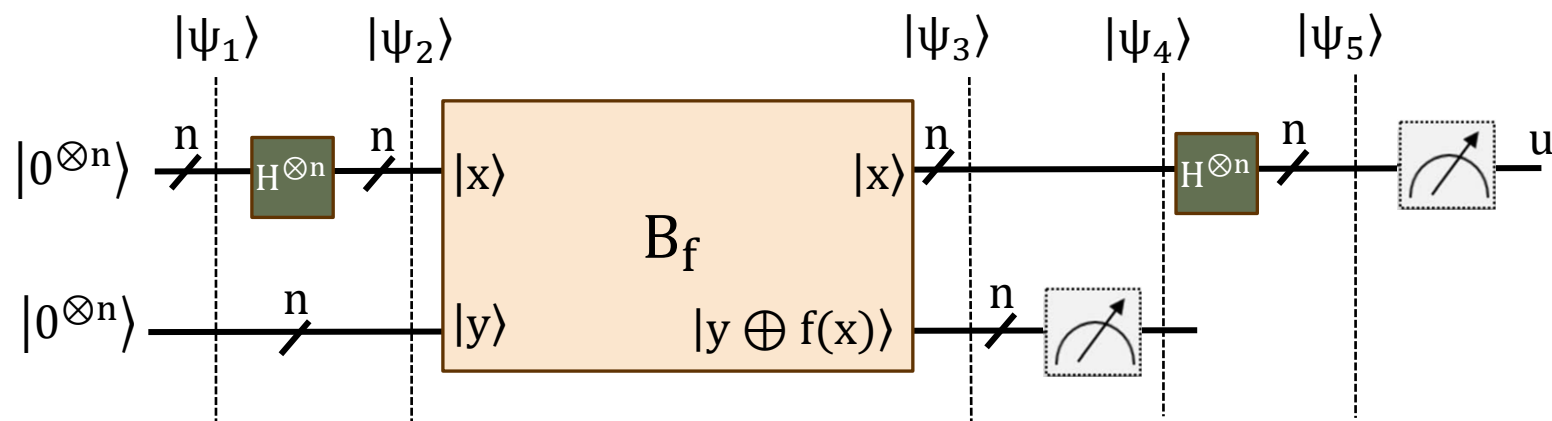
Earlier we had $s_2 = 0$

If $s_1 = s_3 = 0$, then the entire secret string will be $s = 000$. We set it in the very beginning that 's' is non-zero secret string. So, the only option left is $s_1 = s_3 = 1$. In this case, the $s = 101$

In this small example with a 3-bit input string, we were able to determine the secret string using simple logical reasoning and substitution. Since there were only a few variables, the structure of the equations was easy to see directly.

However, for larger input sizes (large n), this approach quickly becomes impractical. We then systematically solve the system of linear equations using Gaussian elimination over $\text{GF}(2)$.

Simon Algorithm - Summary



- Simon's algorithm never accesses the truth table of the function $f(x)$
- The truth table is a classical description used only by us (the designers) to understand the problem
- In the actual algorithm, the function $f(x)$ is provided as a quantum oracle B_f

How the Hidden Structure Is Extracted

- The oracle encodes the promise $f(x) = f(x \oplus s)$ in its action.
- The quantum circuit forces different inputs to interfere.
- This interference produces linear equations of the form $u \cdot s = 0$
- These equations are solved classically, without ever knowing $f(x)$

What the Oracle Really Is

- The oracle is a quantum circuit that implements the transformation $B_f|x\rangle|y\rangle \rightarrow |x\rangle|y \oplus f(x)\rangle$
- The internal logic of the oracle is hidden.
- The algorithm can only:
 - Query the oracle
 - Observe quantum interference in the output

Why the Truth Table Is Irrelevant to the Algorithm

- The quantum circuit:
 - Never stores $f(x)$ explicitly
 - Never compares outputs to look for collisions
 - Never loops over inputs
- All structure in f (including the secret string s) is revealed indirectly through interference.